EXAM SHIELD

Software Engineering Question Bank

Processes, design principles, testing and collaboration.

13 questions: 8 multiple choice, 3 fill in the gap and 2 written. Total 19 marks.
Edition: with answers (staff copy, ready to import into Exam Shield).

How this paper is laid out

Questions are numbered in one sequence. Multiple-choice options are lettered with a full stop; each multiple-choice question is worth one mark unless stated.

Fill-in-the-gap questions show each blank as a line; they earn one mark per blank, shown in square brackets, and are marked automatically.

Written questions show their marks in square brackets and are marked by an examiner.

To use this paper in Exam Shield, open an exam, choose Questions and then Import from PDF, and upload this file. Review the preview and import.

Section 1: Software Engineering

Processes, design principles, testing and collaboration.

1. Which agile framework organises work into fixed-length sprints?

A. Spiral
B. Scrum
C. Waterfall
D. V-model

Answer: B

2. What does unit testing focus on?

A. Only the user interface
B. Only the hardware
C. The smallest testable parts of a program
D. The whole system end to end

Answer: C

3. Which Git command records staged changes in the repository history?

A. git push
B. git commit
C. git clone
D. git status

Answer: B

4. In the SOLID principles, what does the S stand for?

A. Simple design
B. Static typing
C. Single responsibility
D. Secure coding

Answer: C

5. What is refactoring?

A. Rewriting the program in another language
B. Adding new features
C. Improving code structure without changing its behaviour
D. Deleting failing tests

Answer: C

6. What does CI stand for in modern development practice?

A. Code Inspection
B. Central Index
C. Compiled Interface
D. Continuous Integration

Answer: D

7. Which UML diagram shows how objects interact over time?

A. Sequence diagram
B. Class diagram
C. Entity-relationship diagram
D. Deployment diagram

Answer: A

8. What is technical debt?

A. Unpaid developer salaries
B. Loans used to buy hardware
C. Money owed for software licences
D. The future cost of shortcuts taken in code today

Answer: D

9. A Git _____ lets a developer work on a feature in isolation before merging it. [1 mark]

Answer: branch

10. The design pattern that ensures a class has only one instance is the _____ pattern. [1 mark]

Answer: Singleton

11. Re-running tests to check that existing features still work after a change is _____ testing. [1 mark]

Answer: regression

12. Compare the Waterfall and Agile approaches to software development. [5 marks]

13. Explain the purpose of code reviews and two benefits they bring to a team. [3 marks]